Reflections

1 Plot point A on a coordinate plane.
Then, plot point A', a reflection of point A across the x-axis.

a $A\,(7,3)$ **b** $A\,(-5,-6)$ **c** $A\,(2,-2)$ **d** $A\,(-5,6)$

2 Plot point A on a coordinate plane.
Then, plot point A', a reflection of point A across the y-axis.

a $A\,(6,9).$ **b** $A\,(-7,-3)$ **c** $A\,(9,-10)$ **d** $A\,(-7,8)$

3 Consider the following:

a Plot the point $A\,(6,7).$

b If point A is reflected about the x-axis to produce point B, determine the coordinates of point B.

c How many units would point A need to be translated to overlap point B?

4 Consider the point $A\,(6,3).$

a Plot point A on the number plane.

b If point A is reflected across the y-axis to produce point B, determine the coordinates of point B.

c What distance would point A need to be translated to overlap point B?

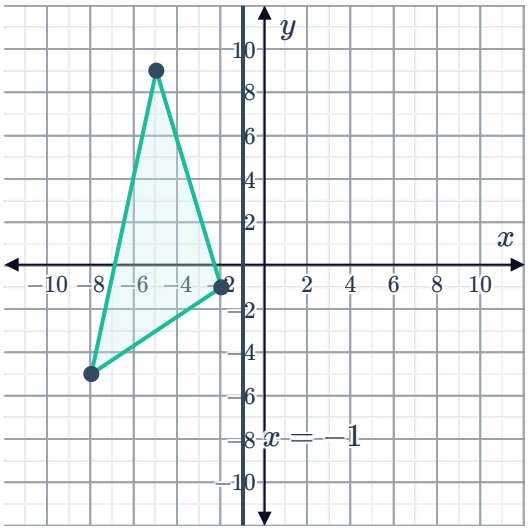
5 Plot the line segment AB owhere the endpoints are $A\,(-4,-2)$ and $B\,(6,7)$. Then plot the reflection of the line segment AB across the x-axis.

6 Plot the line segment AB, where the endpoints are $A\,(-6,-1)$ and $B\,(10,8)$. Then, plot the reflection of the line segment about the y-axis.

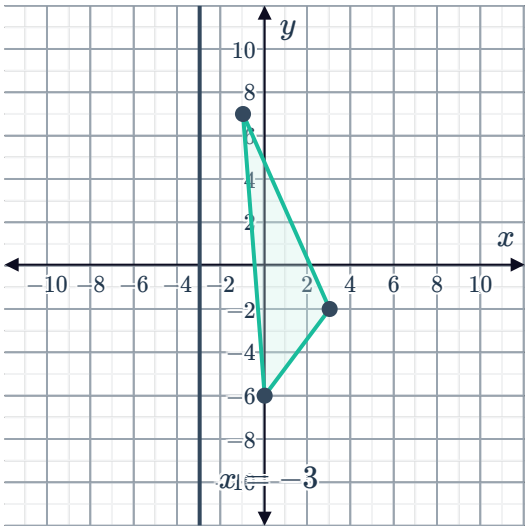
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7 Plot the reflection of each of the following triangles across the given line

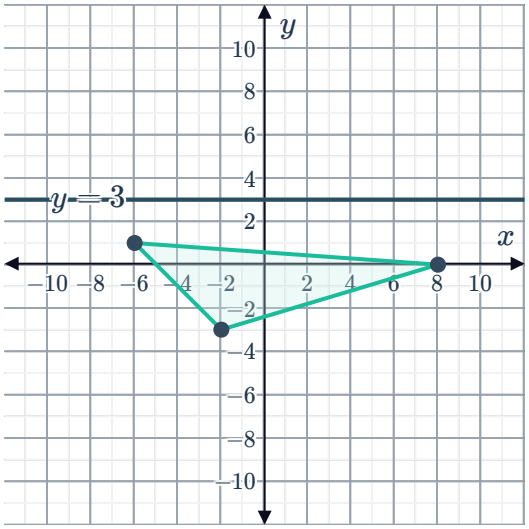
a



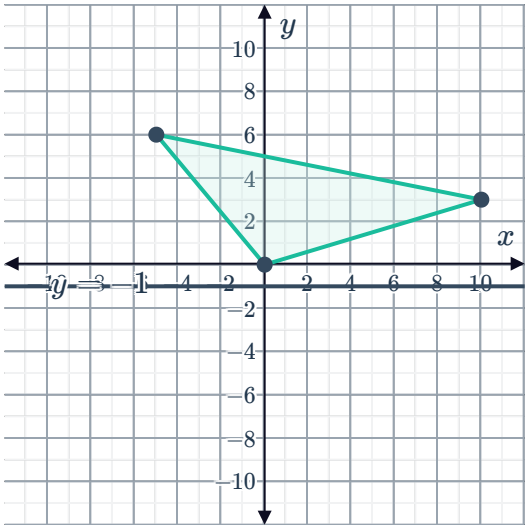
b



c



d



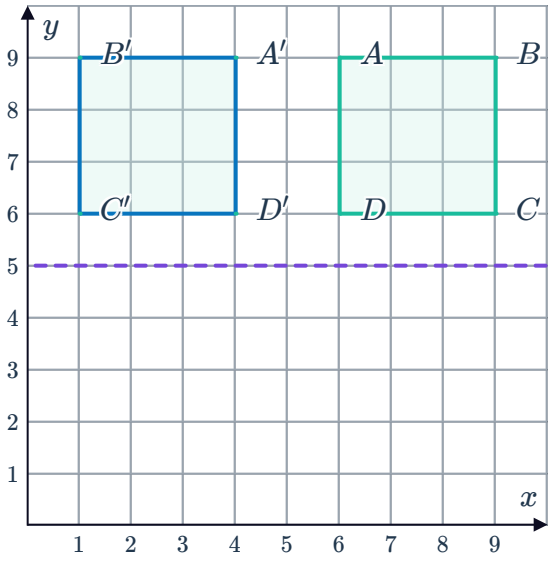
Additional questions

8 Describe the result of reflecting a figure across the x-axis.

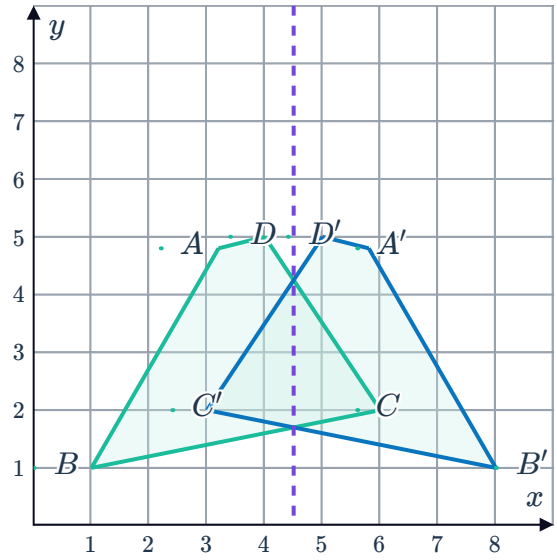
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9 Determine whether each of the following transformations is a reflection across the given line:

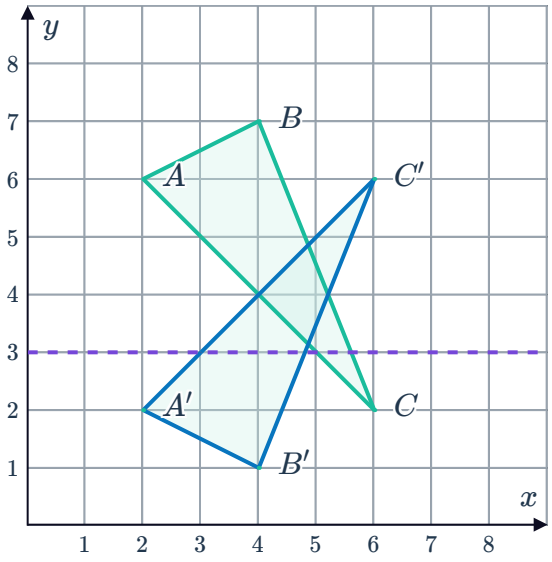
a



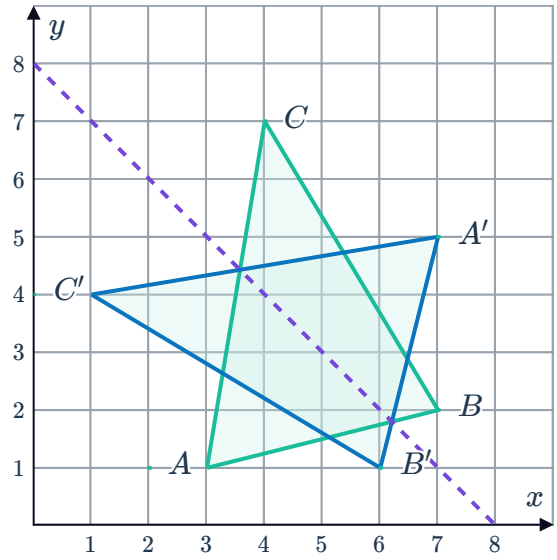
b



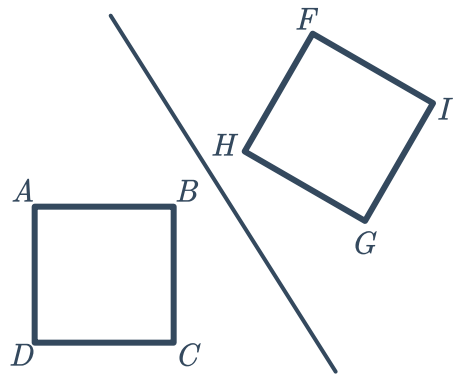
c



d



10 Square $ABCD$ was reflected across the given line. Which point in the image corresponds to point A?

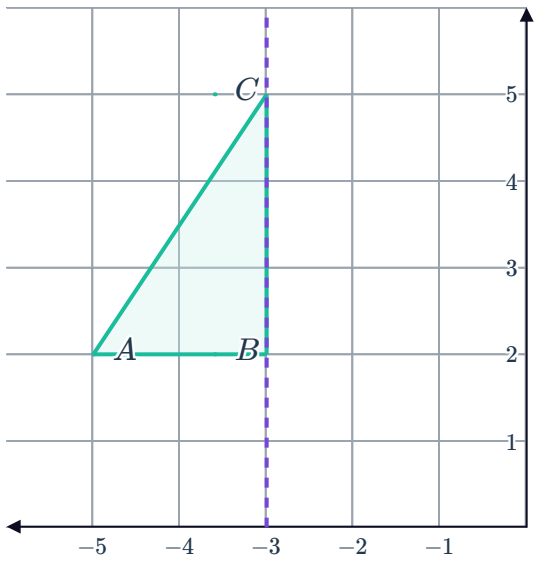


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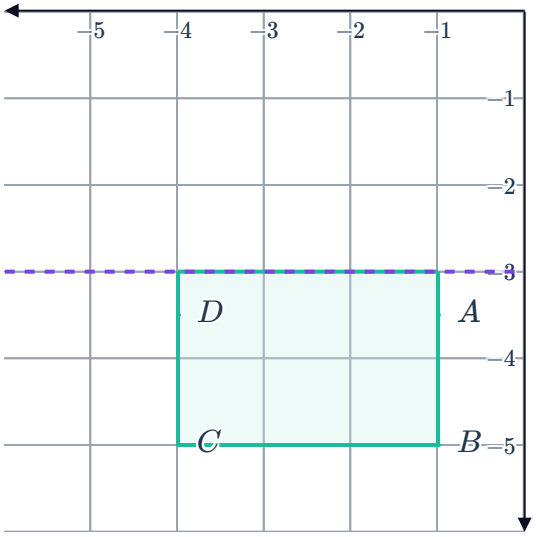
11 Describe the characteristics of a figure that has reflectional symmetry and sketch an example.

12 Reflect each of the following shapes across the given line of reflection. Label all vertices on the image.

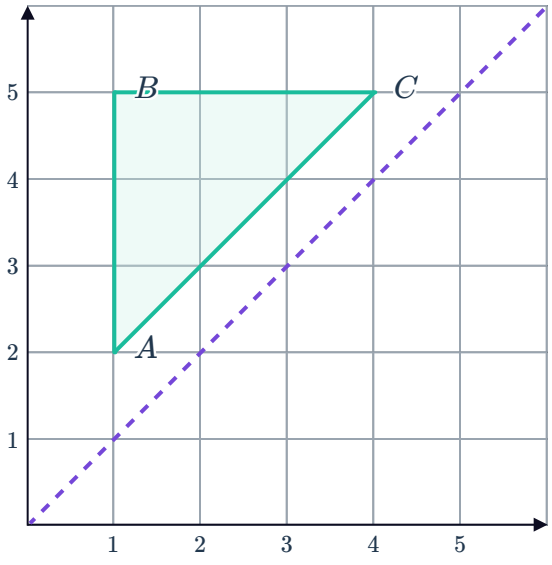
a



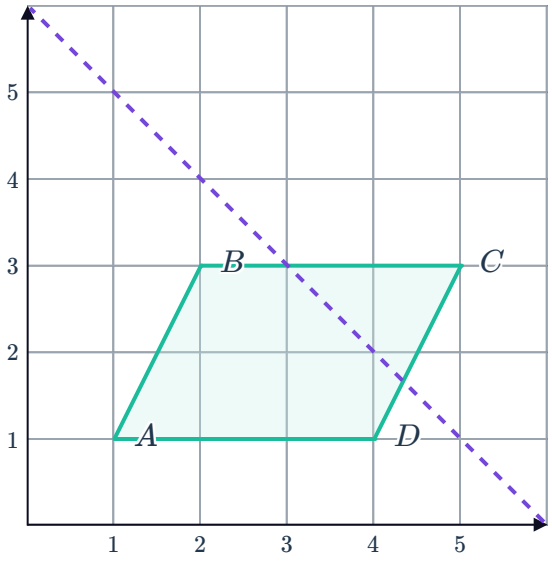
b



c



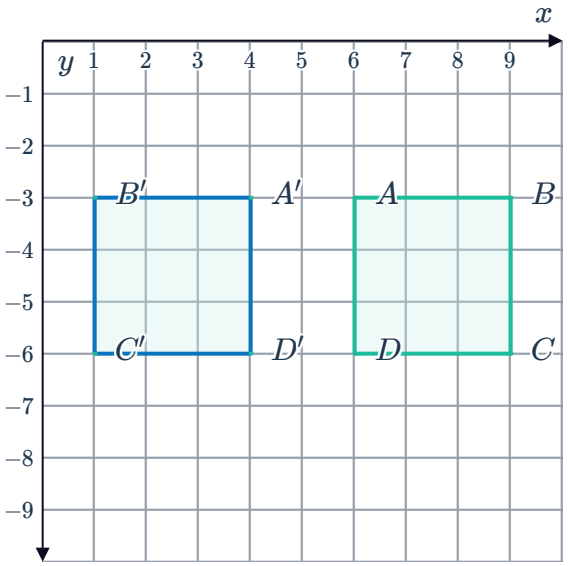
d



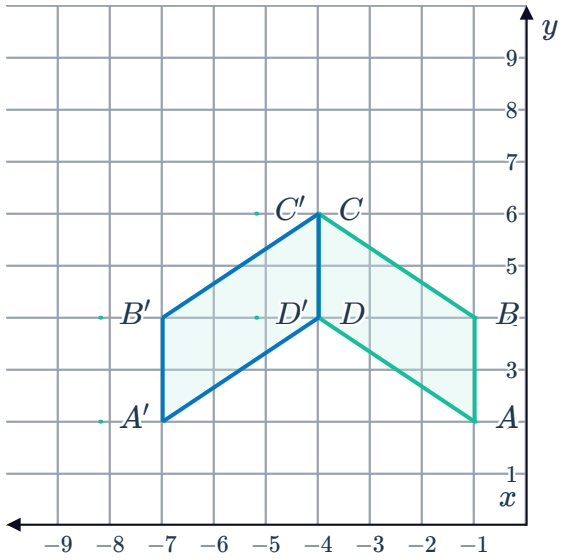
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13 Draw the line of reflection for each of the following transformations:

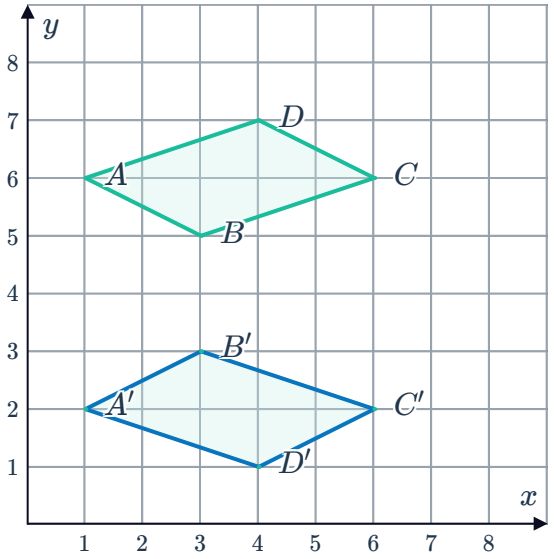
a



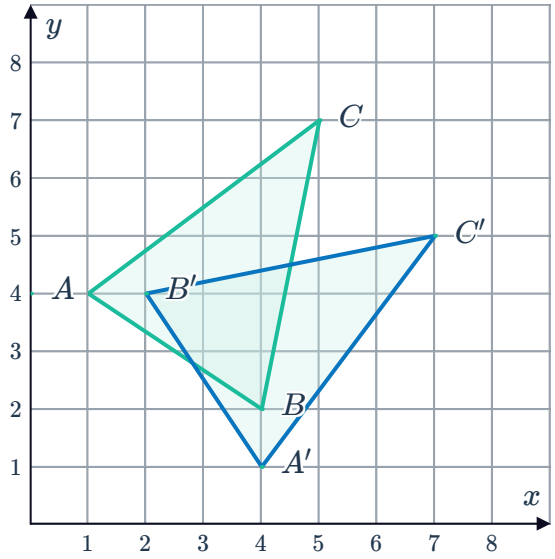
b



c



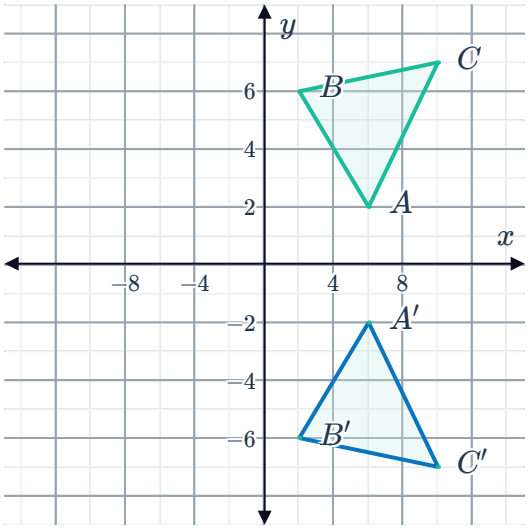
d



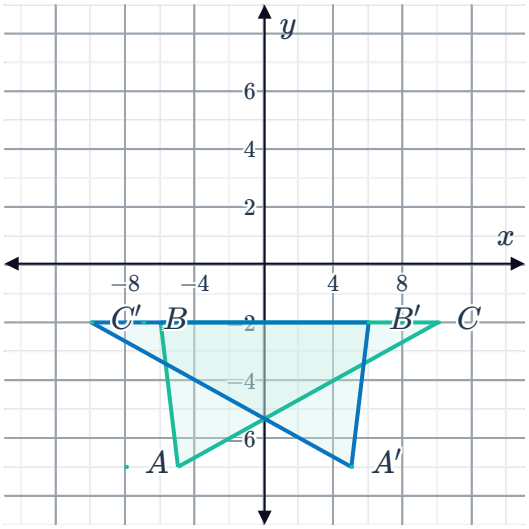
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14 Describe each of the following reflections:

a

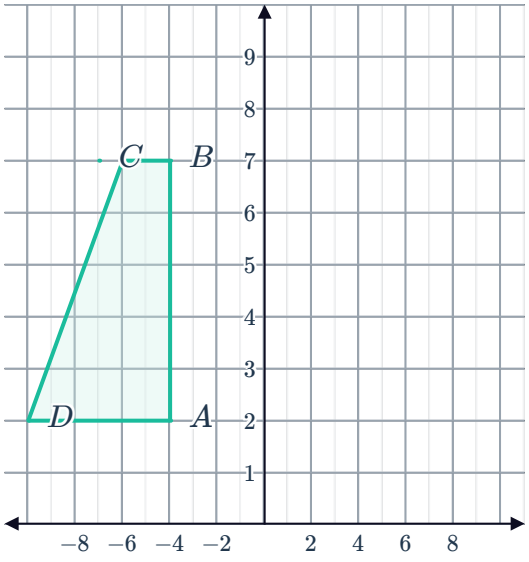


b

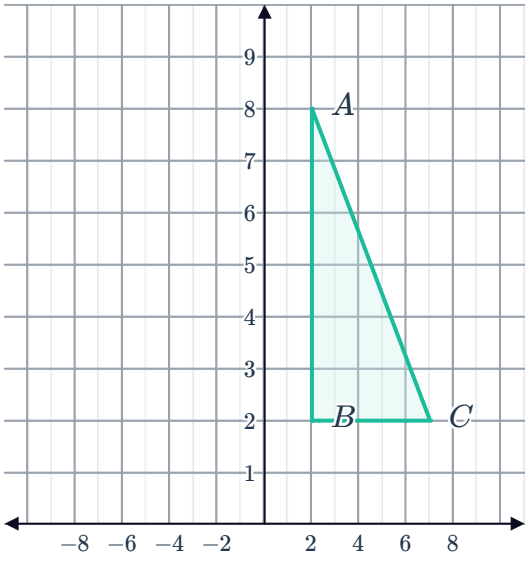


15 Reflect the following shapes across the given the lines of reflection:

a Line of reflection: y-axis

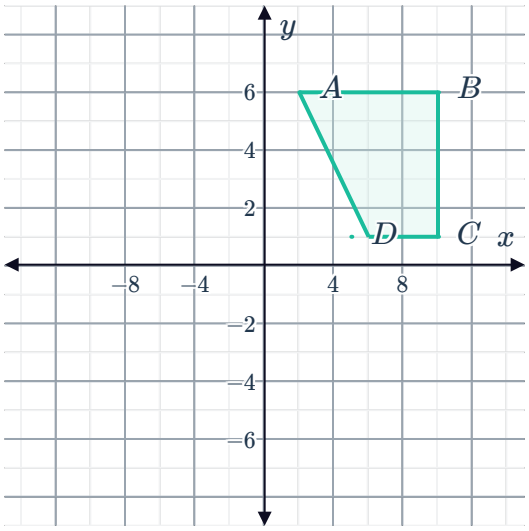


b Line of reflection: y-axis

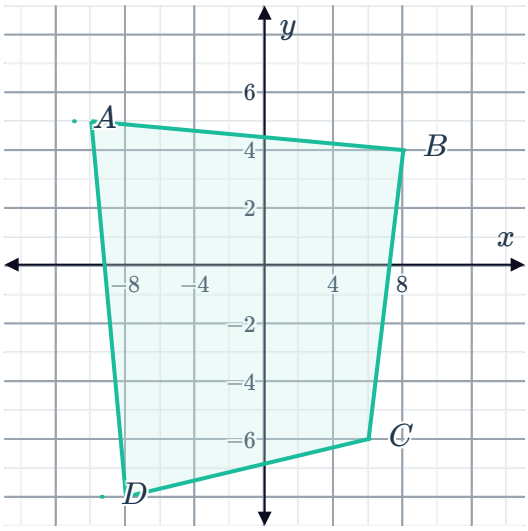


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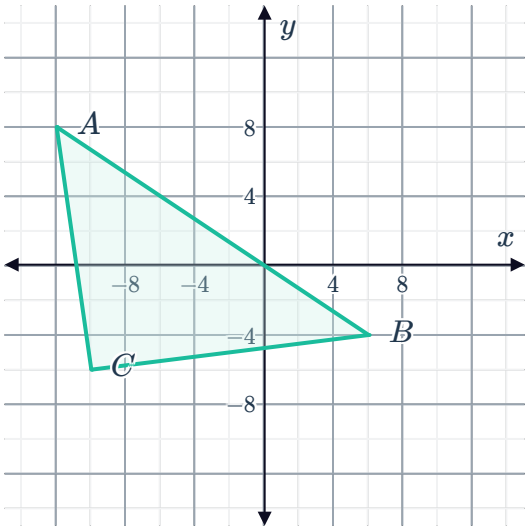
c Line of reflection: x-axis



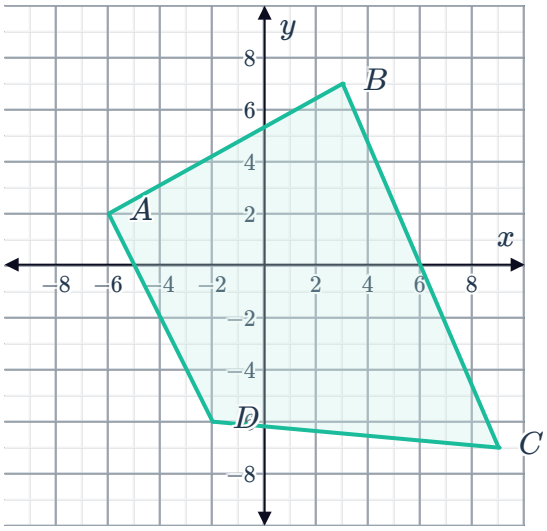
d Line of reflection: x-axis



e Line of reflection: $y = x$



f Line of reflection: $y = -x$



16

a If point $M(5, 6)$ is reflected across the y-axis to produce point N, what are the coordinates of point N?

b What distance would the original point M need to be translated to map to point N?

17 If point $F(-3, -7)$ is reflected across the line $y = -x$ to produce point B, what are the coordinates of point B?

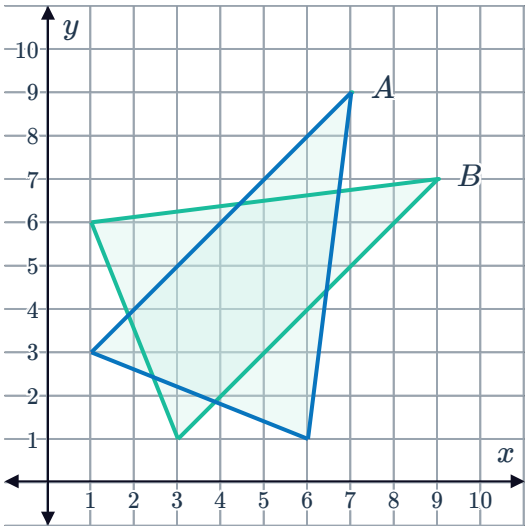
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18 Determine whether each of the following statements are true or false:

- **a** When a figure is reflected across the x-axis, only the sign of the y-value changes.
- **b** When a figure is reflected across a vertical line, only the y-value changes.
- **c** When a figure is reflected, the preimage and image have the same size and shape.
- **d** When a figure is reflected, the preimage and image have the same orientation.
- **e** A figure is reflected across a line. If the reflected figure is then reflected across the same line again, it will return to the original figure.

19 Sokayna drew figure A and Roly drew figure B.

- **a** Can Sokyana's triangle be reflected to produce Roly's? If so, describe the reflection.
- **b** Roly thinks his triangle has a greater perimeter than Sokanya's. Is this true? Explain why or why not.



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